Supplementary Methods: Constrained Bivariate Gaussian Mixture Modeling

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1. Objective and Input Data

To determine whether the mRNA–protein response regimes identified by Differential Pattern Analysis (DiPa) reflect distinct biological subpopulations rather than arbitrary slices of a continuous cloud, we fitted constrained bivariate Gaussian Mixture Models (GMM) to the empirical $\log_2$ fold changes independently of the ± 0.5 cutoffs.

The input is a matrix $X \in \mathbb{R}^{N\times 2}$ ($N = 943$ core matched pairs), where for each gene i:

$$X_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{pmatrix} \log_2 \mathrm{FC}_{\mathrm{RNA},i} \\ \log_2 \mathrm{FC}_{\mathrm{Prot},i} \end{pmatrix}. \tag{1}$$

2. Model Specification and Biophysical Constraints

The probability density of observation X_i under a mixture of $K \in \{1, 2, 3\}$ bivariate Gaussians is:

$$p(X_i \mid \theta) = \sum_{k=1}^{K} \pi_k \mathcal{N}_2 \left(X_i \mid \mu_k, \Sigma_k\right), \quad \text{with } \pi_k \geq 0, \quad \sum_{k=1}^{K} \pi_k = 1, \tag{2}$$

where $\mu_k = (\mu_{xk}, \mu_{yk})^\top$ are component centroids and $\Sigma_k = \begin{pmatrix} \sigma_{xk}^2 & \rho_k \sigma_{xk}\sigma_{yk} \\ \rho_k \sigma_{xk}\sigma_{yk} & \sigma_{yk}^2 \end{pmatrix}$ are covariance matrices.

To prevent unconstrained components from fitting arbitrary noise, three biophysical constraints were enforced:

1. **Component 1 (Diagonal / Co-regulation):** Represents canonical stoichiometric coupling. The correlation is bounded within a biologically realistic coupling window:
$$\rho_1 \in [0.70,\, 0.90]. \tag{3}$$
2. **Component 2 (Vertical / Protein-dominant decoupling):** Represents protein changes occurring without significant transcriptomic shifts. Coordinate correlation is set to zero, and variance is vertically elongated by factor c:
$$\rho_2 = 0, \qquad \sigma_{y2}^2 \geq c \cdot \sigma_{x2}^2. \tag{4}$$
3. **Component 3 (Horizontal / Post-transcriptional buffering, for $K = 3$):** Represents large mRNA changes buffered at the protein level. Correlation is set to zero, and variance is horizontally elongated:
$$\rho_3 = 0, \qquad \sigma_{x3}^2 \geq c \cdot \sigma_{y3}^2. \tag{5}$$

The elongation factor was systematically tested across $c \in \{1.3, 1.5, 2.0, 2.5, 4.0\}$ (aspect ratios $\sqrt{c} \in [1.14, 2.00]$).

3. Expectation-Maximization (EM) Algorithm

Parameters were estimated iteratively via the Expectation-Maximization (EM) algorithm [1, 2] until the log-likelihood converged ($|\ell^{(t+1)} - \ell^{(t)}| < 10^{-6}$):

- **Initialization:** Posterior weights $\gamma_{ik}^{(0)} \in [0, 1]$ were initialized deterministically based on coarse regions (vertical candidates: $|x_i| < 0.55, |y_i| > 0.40$; horizontal candidates: $|x_i| > 0.50, |y_i| < 0.50$; diagonal otherwise), normalized so $\sum_k \gamma_{ik}^{(0)} = 1$.
- **M-Step (Maximization):** Given weights γ_{ik}, component sample sizes $N_k = \sum_i \gamma_{ik}$ and mixing

weights $\pi_k = N_k/N$ were computed. Centroids and sample dispersions were updated:

$$\mu_k = \frac{1}{N_k}\sum_{i=1}^{N}\gamma_{ik}X_i, \quad v_{xk} = \frac{1}{N_k}\sum_{i=1}^{N}\gamma_{ik}(x_i-\mu_{xk})^2, \quad v_{yk} = \frac{1}{N_k}\sum_{i=1}^{N}\gamma_{ik}(y_i-\mu_{yk})^2, \tag{6}$$

and $c_{xy,1} = \frac{1}{N_1}\sum_{i=1}^{N}\gamma_{i1}(x_i-\mu_{x1})(y_i-\mu_{y1})$. Constraints were then directly applied to define the final dispersion parameters $\hat{\sigma}^2$ and matrices Σ_k:

- For $k=1$: $\hat{\sigma}^2_{x1} = v_{x1}$, $\hat{\sigma}^2_{y1} = v_{y1}$, $\hat{\rho}_1 = \min(0.90, \max(0.70, c_{xy,1}/\sqrt{v_{x1}v_{y1}}))$, and $\Sigma_1 = \begin{pmatrix} \hat{\sigma}^2_{x1} & \hat{\rho}_1\hat{\sigma}_{x1}\hat{\sigma}_{y1} \\ \hat{\rho}_1\hat{\sigma}_{x1}\hat{\sigma}_{y1} & \hat{\sigma}^2_{y1} \end{pmatrix}$.
- For $k=2$: $\hat{\sigma}^2_{x2} = v_{x2}$, $\hat{\sigma}^2_{y2} = \max(v_{y2},\, c\cdot v_{x2})$, and $\Sigma_2 = \mathrm{diag}(\hat{\sigma}^2_{x2},\, \hat{\sigma}^2_{y2})$.
- For $k=3$: $\hat{\sigma}^2_{x3} = \max(v_{x3},\, c\cdot v_{y3})$, $\hat{\sigma}^2_{y3} = v_{y3}$, and $\Sigma_3 = \mathrm{diag}(\hat{\sigma}^2_{x3},\, \hat{\sigma}^2_{y3})$.

- **E-Step (Expectation):** Using the updated parameters, posterior responsibilities were recalculated:

$$\gamma_{ik}^{(t+1)} = \frac{\pi_k\,\mathcal{N}_2\left(X_i \mid \mu_k,\, \Sigma_k\right)}{\sum_{j=1}^{K}\pi_j\,\mathcal{N}_2\left(X_i \mid \mu_j,\, \Sigma_j\right)}, \quad \ell^{(t+1)} = \sum_{i=1}^{N}\ln\left(\sum_{k=1}^{K}\pi_k\,\mathcal{N}_2\left(X_i \mid \mu_k,\, \Sigma_k\right)\right). \tag{7}$$

4. Model Selection (BIC)

Model selection across $K \in \{1,2,3\}$ and $c \in \{1.3, 1.5, 2.0, 2.5, 4.0\}$ was evaluated using the Bayesian Information Criterion (BIC) [3]:

$$\mathrm{BIC} = -2\hat{\ell} + P\ln(N), \tag{8}$$

where $\hat{\ell}$ is the maximized log-likelihood, $N = 943$ ($\ln(943) \approx 6.849$), and P is the number of free parameters:

- $K=1$: 2 means + 2 variances + 1 covariance $\implies P = 5$.
- $K=2$ (constrained): 1 weight (π_1) + 4 means + 3 dispersion (Comp 1) + 2 variances (Comp 2) $\implies P = 10$.
- $K=3$ (constrained): 2 weights + 6 means + 3 (Comp 1) + 2 (Comp 2) + 2 (Comp 3) $\implies P = 15$.

A difference $\Delta\mathrm{BIC} < -10$ provides decisive evidence in favor of the lower-BIC model on the Kass–Raftery scale [4]. BIC was chosen over classical likelihood-ratio tests because testing component complexity (K vs. $K+1$) places the null hypothesis ($\pi_{K+1} = 0$) on the boundary of the probability simplex, violating Wilks' regularity conditions [2].

5. Confidence Ellipses

Concentration ellipses for component k at confidence levels $\alpha \in \{0.50, 0.90, 0.95\}$ were generated using the spectral decomposition $\hat{\Sigma}_k = V\Lambda V^\top$:

$$E_k(\alpha) = \hat{\mu}_k + \sqrt{F^{-1}_{\chi^2_2}(\alpha)}\cdot V\Lambda^{1/2}\begin{pmatrix}\cos\phi \\ \sin\phi\end{pmatrix}, \quad \phi \in [0, 2\pi), \tag{9}$$

where V is the orthogonal matrix of eigenvectors governing the spatial orientation (rotation) of the ellipse axes, $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2)$ contains the eigenvalues (variances along principal axes), and $\sqrt{F^{-1}_{\chi^2_2}(\alpha)}$ is the Mahalanobis critical radius derived from the χ^2_2 distribution enclosing probability mass α ($r_{0.50} \approx 1.177$, $r_{0.90} \approx 2.146$, and $r_{0.95} \approx 2.448$).

The complete analysis pipeline is implemented in base R and `data.table` (see script `Generate_Supplementary_Figure_6_Master_Plate.R`).

References

[1] A. P. Dempster, N. M. Laird, and D. B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *J. R. Stat. Soc. Series B*, 39(1):1–22, 1977.

[2] G. J. McLachlan and D. Peel. *Finite Mixture Models*. John Wiley & Sons, New York, 2000.

[3] G. Schwarz. Estimating the dimension of a model. *Ann. Stat.*, 6(2):461–464, 1978.

[4] R. E. Kass and A. E. Raftery. Bayes factors. *J. Am. Stat. Assoc.*, 90(430):773–795, 1995.